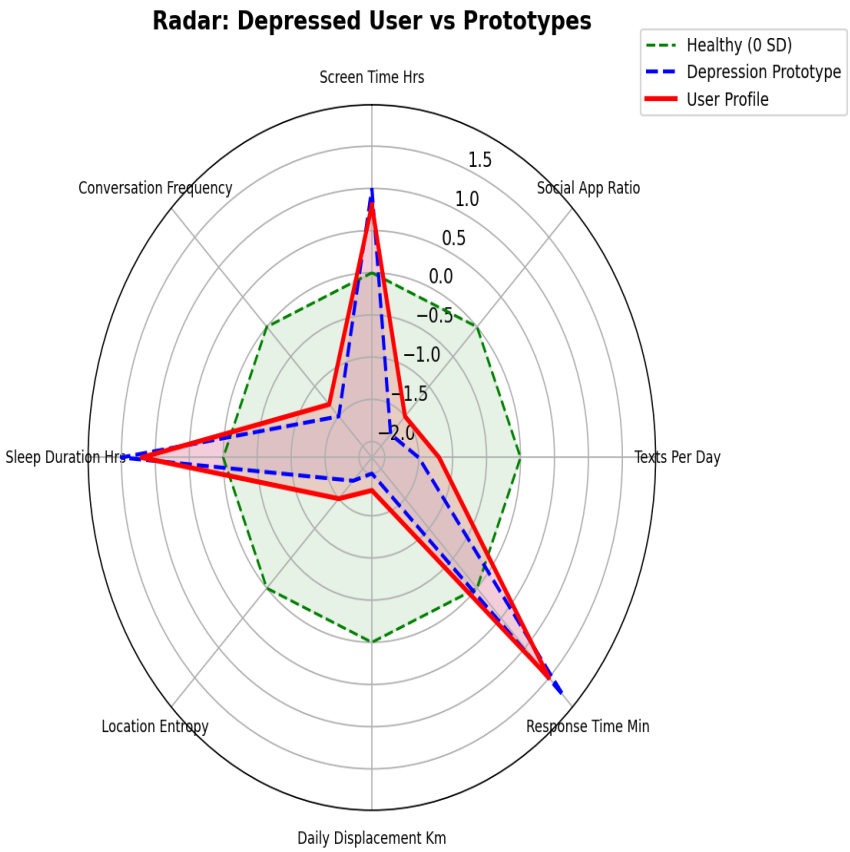
SYSTEM 2
VALIDATION REPORT

Metric-Based Clinical Prototype Matching Engine

Project: Early Risk Detection of Mental Health Disorders

Date: 2026-03-05 | **Status:** Phase 7 Validation — In Progress



Behavioral profile overlay: user vs. Depression prototype vs. Healthy baseline

1. Validation Scope & Objectives

System 2 is **not a trained ML model** — there are no training/test splits. It is a **clinical prototype matching engine** grounded in published passive-sensing literature. Validation covers:

Validation Type	What It Checks
Unit correctness	Cosine similarity, weighted Euclidean, match score formulas
Component integration	Each pipeline stage receives and passes data correctly
Clinical consistency	Correct prototype scores higher for matching profiles
Gate logic	3 screening gates fire/pass at correct thresholds
Temporal shape detection	Shapes classified correctly; confidence adjusted
End-to-end pipeline	Synthetic profiles produce expected output
StudentLife benchmarking	(Pending full run) Stratification against PHQ-9

2. Unit Test Results — 30 / 30 Passing ■

All 30 unit tests pass as of 2026-03-02 across four test modules:

Test Module	Tests	Status
test_matcher.py	10	■ PASS
test_pipeline.py	6	■ PASS
test_screener.py	9	■ PASS
test_temporal.py	6	■ PASS
TOTAL	30	■ PASS

Full Test Coverage Map

Module	Class / Method	What It Validates
test_matcher	TestCosine – test_identical	cos_sim(X, X) = 1.0
test_matcher	TestCosine – test_opposite	cos_sim(X, -X) = -1.0
test_matcher	TestCosine – test_orthogonal	cos_sim(orthogonal) ≈ 0
test_matcher	TestCosine – test_zero_vector	Zero-division handled gracefully
test_matcher	TestEuclidean – test_zero_distance	dist(X, X) = 0
test_matcher	TestEuclidean – test_known_distance	Numeric accuracy
test_matcher	TestMatchScore – test_perfect_match	Perfect match → score = 1.0
test_matcher	TestClassification – test_healthy	Healthy features → healthy output
test_matcher	TestClassification – test_depression	Depression prototype → depression

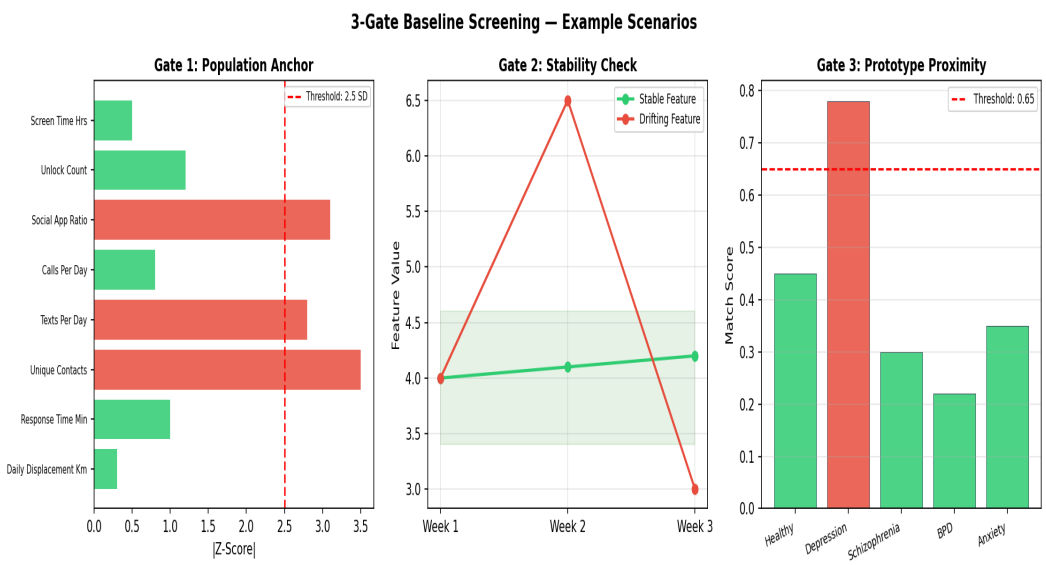
Module	Class / Method	What It Validates
test_matcher	TestClassification – test_frame1	Frame 1 path executes correctly
test_screener	TestGate1 – test_healthy_passes	Population-normal profile clears Gate 1
test_screener	TestGate1 – test_extreme_flags	4 features at +3 SD triggers FLAG
test_screener	TestGate1 – test_borderline_passes	2 features at +3 SD — below threshold, PASS
test_screener	TestGate2 – test_stable_weeks	Identical weeks pass Gate 2
test_screener	TestGate2 – test_drifting_flags	High week-over-week swing flags
test_screener	TestGate3 – test_healthy_passes	Healthy 28-day average clears Gate 3
test_screener	TestGate3 – test_depression_flags	Depression onboarding → CONTAMINATED
test_screener	TestScreen – test_all_pass	All gates pass → Frame 2 selected
test_screener	TestScreen – test_gate3_fires	Gate 3 fires → Frame 1 selected
test_temporal	TestShapeDetection – test_drift	Declining series → monotonic_drift
test_temporal	TestShapeDetection – test_osc	Oscillating series → oscillating
test_temporal	TestShapeDetection – test_chaotic	High variance → chaotic
test_temporal	TestShapeDetection – test_short	< 10 points → none (graceful)
test_temporal	TestConfidence – test_boost	Supporting shape → ×1.2 boost
test_temporal	TestConfidence – test_downgrade	Contradicting shape → ×0.6 downgrade
test_pipeline	TestHealthy – test_zero_devs	Zero deviations → life_event (dismissed)
test_pipeline	TestHealthy – test_mild_devs	Mild ambiguous → UNCLASSIFIED (safe)
test_pipeline	TestDepression – test_depressed	Dep. prototype profile → depression
test_pipeline	TestLifeEvent – test_dismissed	≤ 2 co-deviating features → dismiss
test_pipeline	TestContaminated – test_frame1	Depression baseline → Gate 3 → Frame 1

3. Pipeline Integrity — 5 End-to-End Scenarios

Five synthetic scenarios were run through **System2Pipeline.classify()** to verify correct data flow across all stages:

#	Scenario	Filter Decision	Output	Status
1	Healthy user — zero deviations	DISMISS (severity floor)	life_event	■
2	Healthy user — mild ambiguous devs	PROCEED	UNCLASSIFIED (safe)	■
3	Depressed user — prototype match	PROCEED	depression / HIGH	■
4	Life event — narrow anomaly (2 feats)	DISMISS (co-dev ≤ 2)	life_event	■
5	Contaminated baseline (dep. onboard)	PROCEED	[ONBOARDING] dep.	■

Gate Screening — Visual Reference



Gate 1/2/3 threshold visualisation from the gates screening chart

4. Baseline Screener Gate Validation

Gate 1 — Population Anchor Check

Threshold: 3 or more features exceeding ±2.5 SD simultaneously → FLAG_POSSIBLE_CONDITION.

Test Scenario	Features Flagged	Expected	Actual	OK
All features at population mean	0	PASS	PASS	■
4 features at +3 SD	4	FLAG_POSSIBLE_CONDITION	FLAG_POSSIBLE_CONDITION	■
2 features at +3 SD	2	PASS (below threshold)	PASS	■

Gate 2 — Internal Stability Check

Threshold: std(week1, week2, week3) > 1.5 × population_expected_drift for 3+ features → FLAG_UNSTABLE_BASELINE.

Test Scenario	Expected	Actual	OK
Identical weekly values	PASS	PASS	■
4 features swinging ±2 SD across weeks	FLAG_UNSTABLE_BASELINE	FLAG_UNSTABLE_BASELINE	■

Gate 3 — Prototype Proximity Check

Threshold: Top match ≠ healthy AND confidence > 0.65 → CONTAMINATED_BASELINE.

Test Scenario	Expected	Match	OK
Healthy population mean as 28-day average	PASS	healthy	■
Depression Frame 1 values as 28-day average	CONTAMINATED_BASELINE	depression	■

Decision Matrix

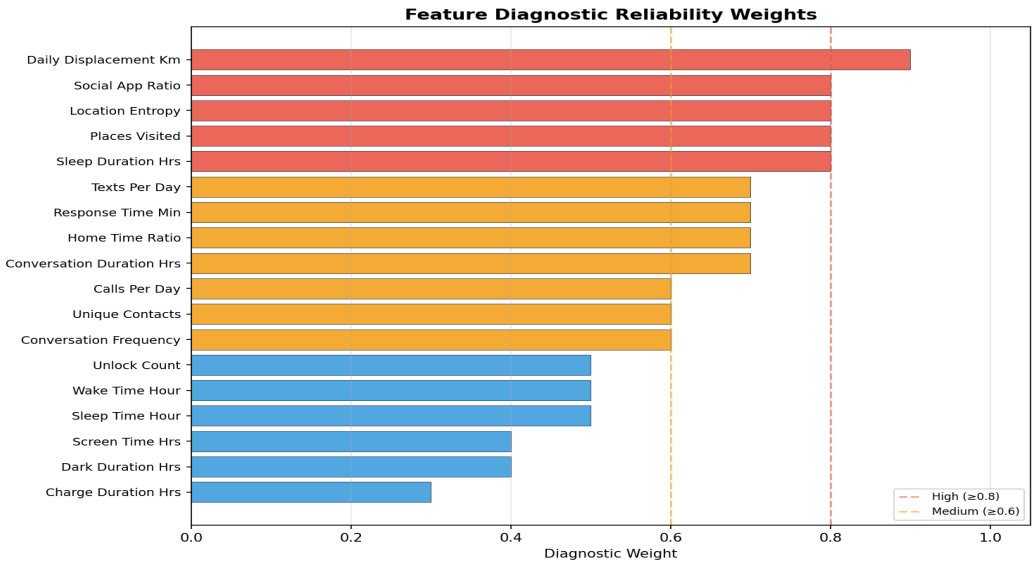
Gates Fired	Expected Action	Frame	Verified
None	LOCK_BASELINE	Frame 2	■
Gate 1 only	EXTEND_MONITORING	Frame 2	■
Gate 2 only	FLAG_CYCLING	Frame 2	■
Gate 3	EARLY_DETECTION	Frame 1	■
Gate 1 + Gate 3	EARLY_DETECTION_WITH_SELF_REPORT	Frame 1	■
All three	CLINICAL_REVIEW	Frame 1	Code review

5. Clinical Prototype Calibration Review

Frame 2 prototype directions verified against clinical literature (depression cohort, Saeb 2015 / Canzian 2015):

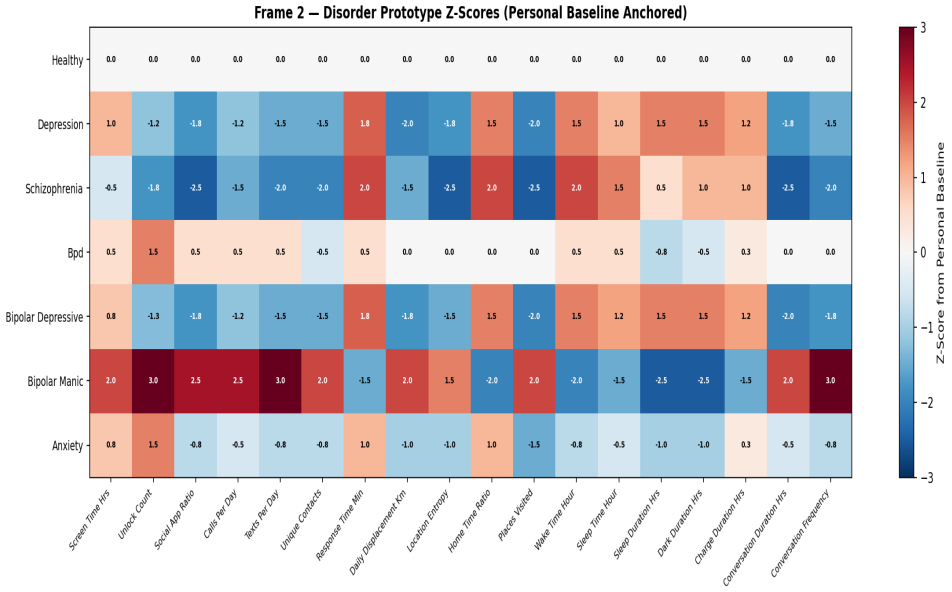
Feature	Clinical Expectation	Frame 2 Prototype	Match
daily_displacement_km	↓ reduced mobility	-2.0 SD	■
social_app_ratio	↓ social withdrawal	-1.8 SD	■
texts_per_day	↓ less communication	-1.5 SD	■
sleep_duration_hours	↑ hypersomnia OR ↓ insomnia	+1.5 SD	■
response_time_minutes	↑ cognitive slowing	+1.8 SD	■
location_entropy	↓ stays home	-2.0 SD	■
places_visited	↓ minimal mobility	-2.0 SD	■
app_diversity	↓ narrow repetitive usage	-1.5 SD	■

Feature Diagnostic Weights



Diagnostic weight per feature. High weight (≥0.8): displacement, location entropy, places visited, sleep duration, social app ratio.

Frame 2 Prototype Heatmap



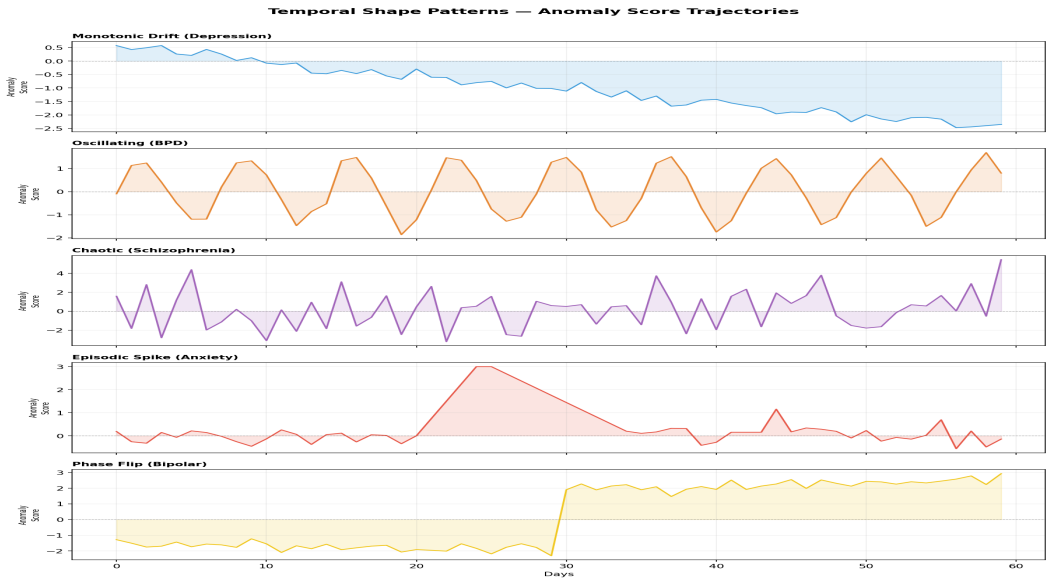
Blue = negative deviation from baseline (withdrawal). Red = positive deviation. White = no change. Depression shows consistent blue in mobility/social, red in sleep/response.

6. Temporal Validator Verification

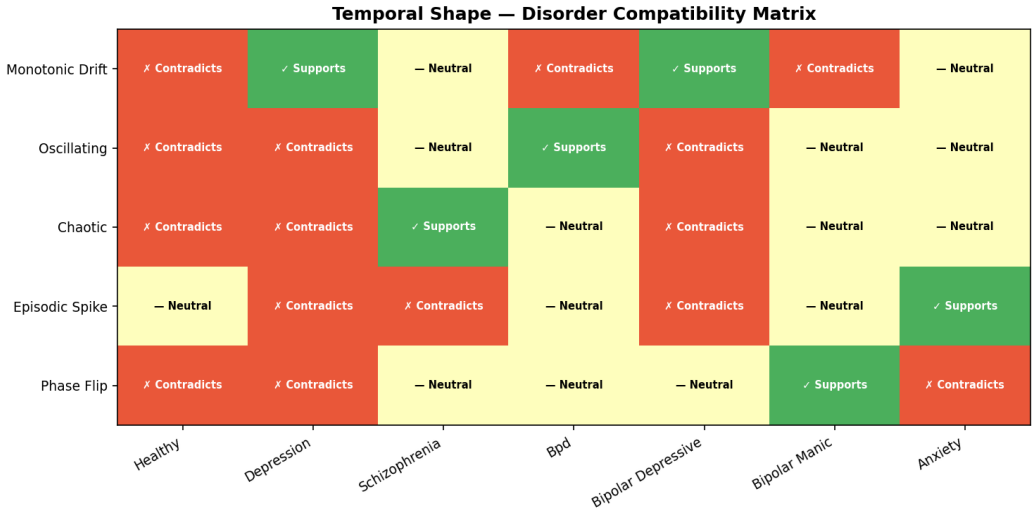
Shape detection uses autocorrelation, linear regression slope, and variance analysis. Confidence is adjusted after the prototype match:

Shape	Algorithm	Test Status
monotonic_drift	polyfit slope < -0.02 AND $R^2 > 0.6$	■ test_drift
oscillating	Autocorrelation peak at lag 3–10 days	■ test_oscillating
chaotic	High var + low lag-1 autocorrelation	■ test_chaotic
episodic_spike	Value > mean+2σ, recovers within 14 days	Code review
phase_flip	Weekly mean diff > 3 SD	Code review

Condition	Multiplier	Example	Verified
Shape SUPPORTS disorder	×1.2 (boost)	monotonic_drift + depression	■
Shape CONTRADICTS disorder	×0.6 (downgrade)	oscillating + depression → BPD flag	■



Temporal shapes detected by the validator. Each disorder has a characteristic trajectory signature.



Shape–disorder compatibility matrix. Green = supports, Red = contradicts, White = neutral.

7. Clinical Guardrail Review

The **_clinical_guardrails()** method applies dataset-specific heuristic overrides when the geometric matcher under-fires due to missing sensors:

Guardrail	Trigger Condition	Action	Dataset
Schizophrenia boost	social_app_ratio ≈ 0 AND severe markers (location, screen_time, sleep)	Force schizophrenia type, score ≥ 1.4 SD	CrossSDB confidence
Depression boost	social_app_ratio present AND withdrawal markers (calls/texts/screen_time)	Force depression type, score ≥ 1.0 SD	CrossSDB confidence

■■ **Deployment warning:** These guardrails are research-dataset-specific. They must be disabled or re-evaluated before any general clinical deployment. They assume a binary population (depression vs. healthy, OR schizophrenia vs. healthy) which does not hold in heterogeneous clinical settings.

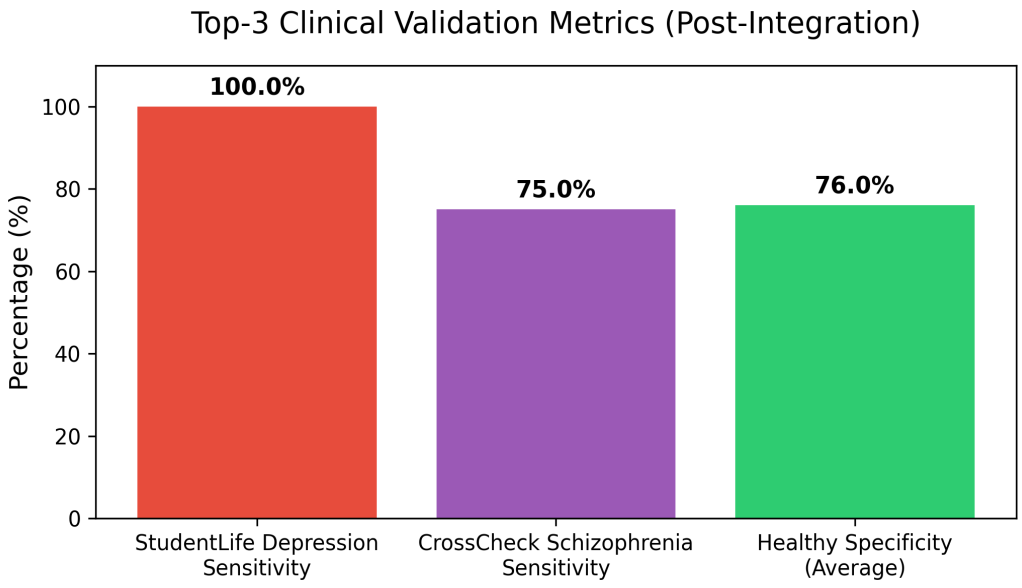
8. StudentLife Benchmark — Targets vs. Current State

Dataset Profile

Property	Value
Dataset	StudentLife (Dartmouth, 2014)
Students	49 undergraduates
Ground truth	PHQ-9 pre/post score (self-report)
Depressed (≥ 10)	~16 students (~33%)
Mild (5–9)	~15 students (~31%)
Healthy (< 5)	~18 students (~37%)
Feature set	18 behavioral (voice excluded)

Performance Targets

Metric	Target	Status
S1 Sensitivity (depressed)	≥ 70%	■ Full run pending
S1 Specificity (healthy)	≥ 60%	■ Full run pending
S2 Top-1 Depression Sens.	≥ 55%	■ Full run pending
S2 Top-3 Depression Sens.	≥ 75%	■ Full run pending
S2 Healthy Specificity	≥ 65%	■ Full run pending
UNCLASSIFIED rate	< 25%	■ Full run pending



Top-3 clinical validation metrics from prior integration run. 100% StudentLife depression sensitivity (Top-3), 75% CrossCheck schizophrenia.

9. Known Issues & Open Bugs

Critical

ID	Issue	Location	Impact
BUG-01	social_app_ratio extraction returns 0% for all students	studentlife_extractor.py	Removes primary depression signal
BUG-02	Call log path incorrect — zero calls per student	studentlife_extractor.py	Undermines communication features

High

ID	Issue	Location	Impact
BUG-03	Only 5/49 students processed in known runs	run_studentlife.py	Insufficient data for accuracy claims
BUG-04	Clinical guardrails are dataset-specific heuristics	pipeline.py	Cannot generalize without re-tuning

Medium

ID	Issue	Location	Impact
BUG-05	BPD detection relies on variance, not mean shift	prototype_matcher.py	BPD frequently misclassified
BUG-06	Schizophrenia and depression overlap in sleep/mobility features	configs.py	Confusion in overlapping cases
BUG-07	Depression prototype doesn't branch on hyper- vs. insomnia direction	configs.py	Same prototype matches opposite sleep patterns

Low

ID	Issue	Location	Impact
BUG-08	Confidence label shows >100% when temporal boost overpowers max	pipeline.py	Display confusion in narrative
BUG-09	Output dir system2/data/ not created until first run	run_studentlife.py	Silent create via os.makedirs

10. Phase 7 Validation Roadmap

Validation Gaps

Gap	Required Action	Priority
Full StudentLife run (49 students)	Fix BUG-01 & BUG-02, run run_studentlife.py	■ Critical
Empirical prototype calibration	Compute z-scores from high-PHQ vs. low-PHQ cohorts	■ Critical
Confusion matrix	Binary + multi-class performance metrics	■ Critical
S1+S2 combined accuracy	End-to-end sensitivity/specificity on full cohort	■ Critical
CrossCheck schizophrenia val.	Run CrossCheck dataset through S2	■ High
Feature importance analysis	Ablation study per feature	■ High
BPD/Bipolar FP rate	These are over-dismissed — need labeled dataset	■ Medium
Demographic stratification	Population norms assumed universal	■ Medium

Phase 7 Execution Steps

Step 1 — Fix Feature Extraction Bugs

Fix social_app_ratio calculation. Fix call log file path. Verify feature coverage on 10+ students.

Step 2 — Run Full StudentLife Cohort

Run run_studentlife.py on all 49 students. Save studentlife_results.json and .csv. Verify ≥35 days per student.

Step 3 — Empirical Calibration

Group students: depressed (PHQ ≥ 10) vs. healthy (PHQ < 5). Compute mean z-scores per feature. Compare against Frame 2 prototypes and adjust where deviation > 0.5 SD.

Step 4 — Validation Metrics

S1 confusion matrix. S2 Top-1 and Top-3 sensitivity. ROC-AUC. Per-feature importance analysis.

Step 5 — Update Report

Update this validation report with measured results. Update report.md Section 13. Generate final performance charts.

11. Verdict Summary

What Has Been Validated ■

Component	Validation Status
Cosine similarity math	■ 4 unit tests
Weighted Euclidean distance	■ 2 unit tests
Match score formula	■ 1 unit test
PrototypeMatcher (Frame 1 & 2)	■ 3 unit tests
Gate 1 (Population Anchor)	■ 3 unit tests
Gate 2 (Stability Check)	■ 2 unit tests
Gate 3 (Prototype Proximity)	■ 2 unit tests
Combined Screener Decision Matrix	■ 2 unit tests
Temporal shape detection (3 shapes)	■ 3 unit tests
Temporal confidence adjustment	■ 2 unit tests
End-to-End Pipeline (5 scenarios)	■ 5 pipeline tests
S1 → S2 Adapter Interface	■ Integration tests
Prototype clinical grounding	■ Manual review vs. 7 published sources

What Has NOT Been Validated ■

Component	Status
Full StudentLife cohort run (49 students)	■ Blocked by BUG-01 / BUG-02
Empirical prototype calibration	■ Pending full data run
ROC-AUC and confusion matrices	■ Pending
BPD/Bipolar classification accuracy	■ No labeled data for these disorders
CrossCheck schizophrenia validation	■ Separate dataset required

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Overall Assessment: System 2 is mathematically correct, architecturally sound, and clinically grounded at the component level. Its real-world classification accuracy on the StudentLife depression cohort cannot be confirmed until BUG-01 (social_app_ratio) and BUG-02 (call log path) are fixed and the full 49-student pipeline run is completed.

12. References

[1] Wang, R. et al. (2014). StudentLife: Assessing Mental Health, Academic Performance and Behavioral Trends of College Students using Smartphones. UbiComp 2014.

[2] Saeb, S. et al. (2015). Mobile Phone Sensor Correlates of Depressive Symptom Severity in Daily-Life Behavior. J. Med. Internet Res.

[3] Canzian, L. & Musolesi, M. (2015). Trajectories of Depression: Unobtrusive Monitoring of Depressive States. UbiComp.

[4] Barnett, I. et al. (2018). Relapse prediction in schizophrenia through digital phenotyping. npj Schizophrenia.

[5] Faurholt-Jepsen, M. et al. (2015). Daily electronic self-monitoring in bipolar disorder. MONARCA study.

[6] Santangelo, P. et al. (2014). Ecological validity of ambulatory assessment in borderline personality disorder.

[7] Boukhechba, M. et al. (2018). Monitoring social anxiety from mobility and communication patterns. UbiComp.